TAP-Path: Task-Adaptive Structural and Token Pruning for Efficient and Reliability-Aware Pathology Foundation Models

Mehedi Hasan[a,*], Ashfak Yeafi[b] and Md Khairul Islam[c]

[a]Department of Computer Science and Engineering, Brac University, Dhaka, Bangladesh
[b]EEE, Khulna University of Engineering & Technology, Khulna, Bangladesh
[c]Mathematics and Computer Science, Hobart and William Smith Colleges, Geneva, NY, USA

ARTICLE INFO

Keywords:
Computational pathology
pathology foundation models
structural pruning
token pruning
trustworthy AI
model efficiency

ABSTRACT

Pathology foundation models (PFMs) provide strong transferable representations, but their large inference footprint can become costly in whole-slide workflows that repeatedly invoke the patch encoder. Existing efficiency strategies mainly reduce adaptation cost, reduce the number of input regions, or replace the parent model through distillation. We propose **TAP-Path**, a task-adaptive framework that instead compresses the pretrained parent itself by jointly reducing transformer depth and token-sequence computation. Starting from Virchow2, **TAP-Path** performs validation-constrained non-contiguous block selection, physically removes unselected blocks, applies input-adaptive late-hierarchy token pruning, and recovers complementary information through a lightweight multi-depth gated head. The selected configuration retains 24 of 32 transformer blocks and 70% of patch tokens after the pruning point, reducing encoder parameters by 24.96% (631.24M to 473.70M) and analytical encoder computation by 35.20% (340.13G to 220.40G FLOPs). Across three task-head seeds, **TAP-Path** achieved 87.98±0.067% accuracy, 81.26±0.49% balanced accuracy, and 82.38±0.48% macro-F1 on a 32-class histopathology benchmark, remaining competitive with full Virchow2 and UNI2-h at substantially lower computational cost. Reliability was assessed through calibration, proper scoring rules, confidence-based failure detection, selective prediction, rare-class sensitivity, and bootstrap uncertainty. The locked model obtained a Brier score of 0.1800±0.0005 and failure-detection AUROC of 0.9047±0.0060. A complementary cross-cohort transfer evaluation on 433 CPTAC whole-slide images yielded 91.22±0.83% accuracy and 91.10±0.81% present-class balanced accuracy. These results show that task-specific redundancy in a large PFM can be removed along both network depth and token dimensions while preserving competitive discrimination and informative reliability characteristics.

1. Introduction

Pathology foundation models (PFMs) have substantially improved representation learning for computational pathology, enabling transfer across image classification, whole-slide analysis, biomarker prediction, retrieval, and prognosis. Models such as UNI, CONCH, Virchow, Prov-GigaPath, CHIEF, Hibou, GPFM, and PathOrchestra have extended this paradigm through larger pretraining cohorts, larger encoders, multimodal supervision, and broader task coverage [9, 24, 25, 28, 36, 38, 41, 42]. As model scale increases, however, computational efficiency becomes an increasingly important part of downstream deployment [2, 20, 45].

This issue is particularly important in digital pathology. Whole-slide images (WSIs) are commonly decomposed into many image regions before slide-level aggregation, so the patch encoder may be executed thousands of times for a single slide. Consequently, a large frozen PFM can remain computationally expensive even when only a small downstream classifier is trained. For example, Virchow2 uses a ViT-H/14-scale encoder with approximately 632M parameters [36]. Freezing the backbone reduces optimization cost but does not remove the resident parameters or most of the forward computation. The capacity useful for broad pretraining may therefore exceed the computation required by a specific downstream pathology task.

Current efficiency strategies address this problem from different directions. Parameter-efficient fine-tuning (PEFT), prompts, and adapters reduce the number of trainable parameters but generally retain the parent backbone during inference [22, 34]. Distillation transfers knowledge to a smaller student, as in SUDA [46], but requires a separate teacher–student optimization stage and changes the deployed model family. Representative-patch selection and multiple-instance learning reduce the number of regions processed or aggregated [40], while lightweight architectures are designed for smaller computational budgets from the outset [15, 18]. These approaches are complementary, but they do not directly answer whether the pretrained parent itself contains task-redundant computation that can be physically removed.

Transformers provide two natural axes for such compression. Structural pruning can remove redundant internal components [1, 27], whereas DynamicViT, EViT, A-ViT, Token Merging, and related approaches reduce computation by shortening or reorganizing the token sequence [4, 5, 21, 23, 33, 39, 44]. Histopathology introduces an additional constraint: diagnostically relevant evidence can be spatially sparse, morphologically subtle, and distributed across scales. Aggressive pruning may therefore remove rare but informative tissue patterns. A useful pathology

*Corresponding author
mehedi.hasan1@g.bracu.ac.bd (M. Hasan)
ORCID(s):

compression strategy must reduce computation without assuming that all low-importance structure or tokens are safely expendable.

Compression should also be evaluated beyond top-line accuracy. Altering the inference graph can change calibration, confidence-based error ranking, and minority-class sensitivity even when discrimination is preserved. PFM benchmarks have reported substantial task- and cohort-dependent variation [3, 7, 19, 29], while pathology robustness studies have highlighted scanner, laboratory, and cohort effects [14, 16, 17]. Calibration, proper scoring rules, failure detection, and selective prediction therefore provide complementary evidence about the behavior of a compressed model [12, 13, 32].

To address these issues, we propose **TAP-Path**, a task-adaptive structural and token pruning framework that compresses a pretrained PFM in place. Starting from Virchow2, **TAP-Path** profiles representational change within the target-domain development data, constructs structurally compressed candidates, and selects a non-contiguous subnetwork using validation performance under explicit compression constraints. The selected encoder physically retains 24 of 32 transformer blocks. Input-adaptive token pruning is then introduced in the later hierarchy, where 70% of patch tokens are retained. Four intermediate representations from the compressed hierarchy are projected and combined through a lightweight learned gate before classification.

The main contributions are:

- **Task-adaptive structural compression of the pretrained parent.** We introduce a target-domain block-novelty screen followed by validation-constrained subnetwork selection and physical reconstruction of a non-contiguous Virchow2 hierarchy, without training a separate student model.
- **Coupled depth–token sparsification with representation recovery.** We combine structural pruning with input-adaptive late-hierarchy token reduction and gated multi-depth feature recovery, explicitly accounting for the interaction between retained depth and token sparsity.
- **Executable accuracy–efficiency characterization.** The selected encoder reduces Virchow2 parameters by 24.96% and analytical FLOPs by 35.20%, with measured runtime profiling, while maintaining competitive performance on the 32-class histopathology task.
- **Reliability-oriented characterization after compression.** We evaluate calibration, proper scoring rules, confidence-based failure detection, selective prediction, rare-class sensitivity, bootstrap uncertainty, and complementary CPTAC cross-cohort transfer to determine whether the efficiency gains preserve informative predictive behavior.

2. Related Work

2.1. Pathology foundation models and downstream adaptation

The progression from pathology-specific transformers to PFMs provides the context for task-adaptive compression. CTransPath used unsupervised contrastive learning to learn transferable histopathology features, while HIPT introduced hierarchical self-supervision for gigapixel pathology images [8, 37]. The subsequent generation of PFMs scaled both data and model capacity. UNI established a general-purpose self-supervised pathology encoder [9]; CONCH paired images and language for visual–language representation learning [24]; Virchow used million-slide-scale pretraining and a ViT-H backbone [36]; Prov-GigaPath introduced a whole-slide hierarchy trained on very large tile collections [41]; CHIEF connected patch and slide representations for diagnosis and prognosis [38]; and Hibou provided open DINOv2-based pathology encoders [28, 31]. GPFM, BEPH, and PathOrchestra further broadened pretraining objectives and downstream task coverage [25, 42, 43]. Recent reviews emphasize that adaptation strategy, dataset composition, evaluation design, and computational scalability are now central to PFM research rather than secondary implementation details [20, 45].

Independent benchmarking has also weakened the assumption that a single PFM should dominate all tasks. Campanella et al. compared public self-supervised PFMs on clinical tasks [7]; Lee et al. showed that adaptation strategy and data regime affect the preferred model [19]; Neidlinger et al. evaluated a broad set of foundation encoders across external cohorts [29]; and a 2026 benchmark further demonstrated task-dependent behavior across general and pathology-specific models [3]. Similarly, Xu et al. studied how foundation representations interact with multiple-instance learning in WSI analysis [40]. These findings motivate same-task comparison and independent transfer evaluation under a fixed downstream protocol rather than relying on headline results from unrelated benchmarks.

2.2. Efficient adaptation, input reduction, and distillation

Efficiency in medical imaging can refer to different resources, and the distinction matters. PEFT, prompts, and adapters reduce the number of parameters optimized during adaptation but generally retain the backbone at inference. Few-shot PEFT has been studied for volumetric medical segmentation [34], while PAMT combines representative patch sampling, visual prompts, and lightweight adaptation for pathology classification [22]. These methods are attractive when optimization cost is the bottleneck but do not necessarily reduce the resident size or forward path of a 600M-parameter encoder.

Distillation addresses inference cost more directly by replacing a large teacher with a compact student. GPFM incorporates expert/self-distillation during foundation-model pretraining [25], and SUDA combines unsupervised distillation with domain adaptation for efficient pathological image

analysis [46]. Input-side methods and MIL pipelines can reduce the number of patches that are aggregated or processed downstream [40]. Lightweight architectures, including efficient transformer/CNN hybrids and histopathology-specific compact designs, represent a further strategy [15, 18, 30]. TAP-Path is complementary to these approaches: it asks whether the pretrained parent itself contains task-redundant computation that can be physically removed while preserving the original representation family.

2.3. Structural sparsification of transformers

Transformers can contain redundancy at the level of heads, channels, layers, and blocks. Early evidence in transformer models showed that many attention heads can be pruned with limited degradation [27]. In vision, structured exploration and re-optimization methods have subsequently searched for sparse subnetworks rather than applying uniform width or depth reduction [1]. Histopathology-specific evidence is beginning to emerge: Boudissa et al. analyzed attention-head redundancy in histology ViTs and reported that selective head pruning can preserve useful classification performance while reducing computation [6]. These studies motivate the hypothesis that redundancy may also exist across the transformer depth of a large pathology foundation encoder.

The unit of structural compression in TAP-Path differs from attention-head pruning. We select and physically instantiate a non-contiguous subset of complete pretrained transformer blocks while keeping hidden dimensionality unchanged. The resulting question is not whether individual heads are redundant, but whether the target task requires the full sequence of transformations learned during broad pretraining. A contiguous-depth baseline is therefore important: if a task-adaptive non-contiguous subnetwork outperforms a naive 24-block truncation at the same parameter count, depth itself is an inadequate proxy for downstream relevance.

2.4. Dynamic token reduction

Token sparsity provides a second, complementary route to efficient transformers. DynamicViT progressively predicts and removes less informative tokens [33]; EViT reorganizes tokens using class-token attention [21]; A-ViT adapts token computation depth without introducing a separate token-routing network [44]; and Token Merging reduces sequence length by combining similar tokens rather than simply discarding them [5]. More recent methods have revisited token pruning for dense prediction, graph-based propagation, and task-aware vision settings [4, 23, 26, 39]. Collectively, this literature establishes that sequence length is often over-provisioned in generic ViTs.

Pathology introduces a stronger preservation constraint. A small glandular pattern, nuclear feature, stromal interface, or rare morphological cue can occupy only a fraction of the image while remaining diagnostically relevant. Multiscale pathology further complicates the assumption that low-attention regions are safely removable. TAP-Path therefore does not treat token reduction as an independent efficiency knob. Tokens are pruned only after stable early representations have formed, the retained ratio is selected on validation data after structural locking, and rare-class behavior is evaluated explicitly. The observed interaction between 24-block and 22-block structures further supports joint selection of structural and token sparsity.

2.5. Reliability and domain robustness

A compressed model may preserve discrimination while changing the quality of its predictive probabilities. Temperature scaling and expected calibration error (ECE) are commonly used to assess calibration [13], whereas negative log-likelihood (NLL) and Brier score evaluate the full predictive distribution. Under dataset shift, uncertainty estimates can deteriorate despite strong in-domain performance [32]. Selective prediction provides a complementary assessment by measuring prediction risk as low-confidence samples are withheld [12].

These considerations are particularly relevant to computational pathology because staining, scanner hardware, tissue preparation, and cohort composition can vary across institutions. Domain-generalization studies emphasize transparent reporting and evaluation under distribution shift [16]; knowledge-guided adaptation has been investigated for cross-domain generalization and demographic fairness [14]; and recent work has shown that PFMs may encode scanner- or laboratory-associated signatures [17]. The present study therefore evaluates calibration, proper scoring rules, confidence-based failure detection, selective prediction, rare-class sensitivity, and statistical uncertainty in addition to predictive performance and computational cost.

Existing studies have addressed PFM adaptation, input reduction, student-model distillation, structural pruning, token sparsification, and reliability evaluation largely as separate problems. The remaining methodological question is whether a large pretrained pathology encoder can be physically reduced along both depth and token dimensions for a target task without replacing the parent model, while retaining useful predictive and confidence behavior. TAP-Path is designed to address this question. Table 1 summarizes the relationship to representative prior work.

3. Materials and Methods

3.1. Study design

The study followed a sequential model-selection and locked-evaluation protocol. Architecture screening and token-retention selection were performed exclusively using the training and validation partitions. The internal test set was not used for transformer-block selection, token-retention selection, loss-function selection, or calibration-hyperparameter optimization. After fixing the structural and token configuration, three independent task heads were trained using seeds 42, 123, and 2026. The CPTAC transfer evaluation was subsequently performed using the locked configuration without model adaptation or threshold

Table 1
Methodological comparison of representative pathology foundation-model adaptation and efficiency strategies. The columns distinguish parameter-efficient adaptation, separate student-model distillation, physical backbone reduction, token/input reduction, reliability analysis, and cross-cohort evaluation. A checkmark denotes an explicitly evaluated component; "–" denotes that the component is not central to the cited method.

Study	Pathology FM	PEFT/adaptation	Separate student	Physical structural reduction	Dynamic token reduction	Reliability analysis	Cross-cohort evaluation
UNI [9]	✓	–	–	–	–	partial	✓
Virchow [36]	✓	–	–	–	–	partial	✓
Lee et al. [19]	✓	✓	–	–	–	–	✓
Xu et al. [40]	✓	–	–	–	input/MIL	–	✓
PAMT [22]	✓	✓	–	–	input sampling	–	✓
SUDA [46]	✓	adaptation	✓	student model	–	–	✓
Boudissa et al. [6]	pathology ViT	–	–	attention heads	–	–	–
DynamicViT / EViT / A-ViT [21, 33, 44]	–	–	–	–	✓	–	–
Token Merging [5]	–	–	–	–	merge	–	–
TAP-Path	✓	task head	–	✓block removal	✓	✓	✓

optimization. This protocol maintains a strict separation between model development, internal testing, and cross-cohort transfer assessment.

The internal benchmark comprised 25,495 image-level records representing 32 cancer classes derived from The Cancer Genome Atlas (TCGA) [35]. The dataset was partitioned into 17,769 training, 3,867 validation, and 3,859 test images, with each image represented using the final 12-patch protocol. Table 2 summarizes the resulting data partitions.

As a complementary cross-cohort transfer assessment, we used 433 whole-slide images (WSIs) from the Clinical Proteomic Tumor Analysis Consortium (CPTAC), comprising 209 clear cell renal cell carcinoma (CCRCC) images [10] and 224 uterine corpus endometrial carcinoma (UCEC) images [11]. The locked model was applied at the image level without CPTAC-specific fitting or threshold optimization. Because these cohorts represent only two of the 32 cancer classes in the internal benchmark, balanced accuracy was computed only over the classes represented in CPTAC. The CPTAC experiment is therefore interpreted as limited cross-cohort transfer evidence rather than broad pan-cancer external validation.

Table 2: Dataset composition used for model development, internal testing, and cross-cohort transfer assessment. Structural and token selection used only the training and validation partitions. The internal test set and CPTAC cohorts were evaluated after configuration locking. CPTAC counts refer to whole-slide images rather than verified unique patients.

Partition	Images	Classes
Training	17,769	32
Validation	3,867	32
Internal test	3,859	32
Internal total	25,495	32
CPTAC transfer	433	2 represented classes

3.2. Preprocessing and patch selection

The internal representation is image-level but is deliberately constructed from multiple pathology regions. The final protocol uses a fixed 12-patch bag for every record. Patch selection was completed before TAP-Path architecture search and therefore does not use test-set model outcomes. Candidate regions were generated at multiple source crop sizes (224, 448, and 896 pixels) and subsequently resized to the encoder input resolution. To preserve scale coverage, the selector first reserves one high-confidence candidate from every available scale. Two additional high-entropy candidates are then forced into the bag so that difficult or ambiguous morphology is not systematically discarded. Remaining positions are filled by a combined utility criterion

$$q_i = 0.50\, c_i + 0.30\, d_i + 0.20\, h_i, \tag{1}$$

The probabilities used by this selector are generated by a reproducible *patch probe*, not by TAP-Path itself. Specifically, Hibou-B patch embeddings are L2-normalized and a multinomial logistic-regression classifier is fitted on training patches only using the LBFGS solver, L2 penalty, $C = 1.0$, and the fixed screening seed. For candidate patch i, if $\mathbf{p}_i^{(q)} \in [0,1]^{32}$ denotes the probe posterior, confidence and normalized predictive entropy are

$$c_i = \max_k p_{ik}^{(q)}, \tag{2}$$

$$h_i = -\frac{1}{\log C} \sum_{k=1}^{C} p_{ik}^{(q)} \log\left(p_{ik}^{(q)} + \epsilon\right), \qquad C = 32. \tag{3}$$

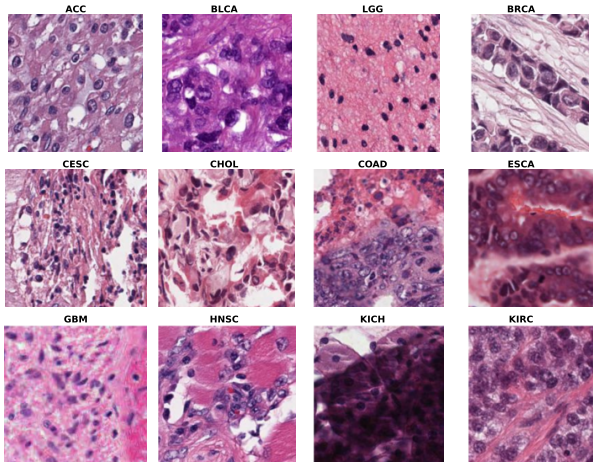
Thus, the probe is a linear probabilistic readout of frozen Hibou-B features; it is neither a ResNet nor a zero-shot classifier. It is trained exclusively on the training split and is used only to construct the deterministic patch manifest.

In Eq. (1), c_i is the probe confidence from Eq. (2), d_i is the minimum cosine distance between candidate i and the already selected Hibou-B patch representations, and h_i is the normalized entropy from Eq. (3). The selection procedure first reserves one highest-confidence patch from every available scale, then forces the two highest-entropy remaining patches, and finally fills the bag according to Eq. (1). The diversity term prevents collapse to near-duplicate tissue regions, whereas the entropy term deliberately preserves difficult regions.

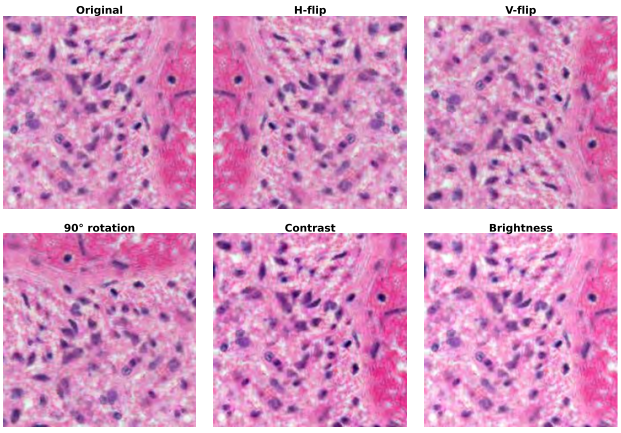
For source image I, stored crop center (x_i, y_i), and source crop size s_i, the selected region is

$$P_i = I\left[y_i - \frac{s_i}{2} : y_i + \frac{s_i}{2}, x_i - \frac{s_i}{2} : x_i + \frac{s_i}{2}\right]. \tag{4}$$

Regions extending beyond the source boundary are padded with white background and resized to 224×224 pixels



(a) Representative internal H&E patches. TCGA abbreviations identify the sampled cancer classes.



(b) Representative label-preserving transformations used during training.

Figure 1: Representative pathology regions and label-preserving training transformations. (a) H&E regions sampled from selected TCGA cancer classes illustrate variation in tissue morphology across the 32-class task. (b) Horizontal/vertical reflection, 90° rotation, and mild contrast/brightness perturbations were applied during training. Validation, internal test, and CPTAC transfer evaluation used deterministic unaugmented regions.

using bicubic interpolation. The resulting 12 regions are deterministic for a given manifest and are reused across compared encoders, preventing model-specific patch selection from contaminating the benchmark. Representative selected regions are shown in Fig. 1.

Data-integrity checks verify the canonical class mapping, finite feature values, feature/index row agreement, deterministic split membership, and absence of zero-vector embeddings. Class imbalance is substantial, motivating balanced accuracy, macro-F1, rare-class analysis, and class-wise reliability rather than relying exclusively on top-1 accuracy. Figure 2 summarizes the long-tailed training distribution across the 32 cancer classes.

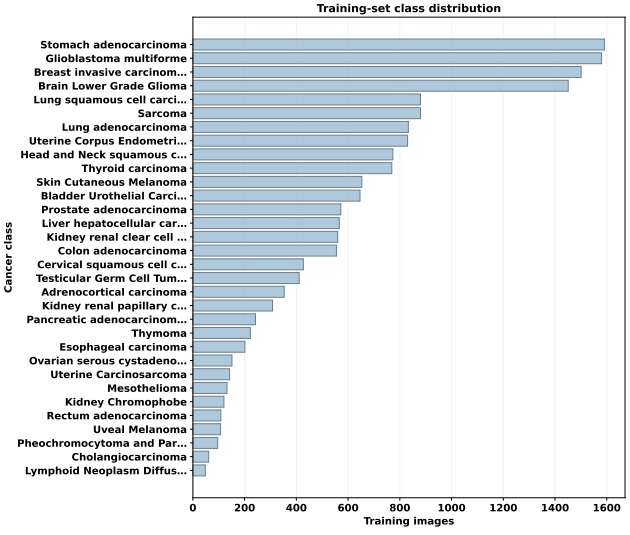


Figure 2: Training-set class distribution for the 32-class TCGA histopathology benchmark. The pronounced long-tailed distribution motivates reporting balanced accuracy and macro-F1 in addition to overall accuracy and supports a separate analysis of classes in the lower quartile of training frequency.

3.3. Data augmentation

Augmentation is restricted to transformations expected to preserve diagnostic class: horizontal/vertical reflection, 90° rotation, mild contrast perturbation, and mild brightness perturbation. An augmented patch is

$$\tilde{P} = \mathcal{T}_\theta(P), \qquad \mathcal{T}_\theta \in \{\mathcal{F}_h, \mathcal{F}_v, \mathcal{R}_{90}, \mathcal{C}_{\delta_c}, \mathcal{B}_{\delta_b}\}. \tag{5}$$

Aggressive color remapping is avoided because stain-dependent appearance can carry diagnostic information. Validation, internal test, and CPTAC transfer evaluation use deterministic, unaugmented patches.

3.4. Foundation-model baselines

Four pathology foundation encoders were retained as interpretable reference points: Hibou-B, CONCH, Virchow2, and UNI2-h. Hibou-B and CONCH represent smaller pathology-specific encoders, while Virchow2 and UNI2-h represent the high-capacity regime. Comparisons use the same 32-class task and dataset partitions from the established benchmark. We report each baseline's measured downstream predictive metrics together with the parameter count and a consistently defined analytical encoder-FLOP estimate. The proposed method is initialized from Virchow2 because it provided a strong large-model baseline while exposing sufficient depth for structural analysis. For additional context, the study evaluated high-compute fusion systems. Virchow2+StaticTriFusion reached 87.43% accuracy with 81.78% macro-F1, and UNI2-h+DenseTriGate reached 87.91% accuracy with 82.68% macro-F1, but both execute three foundation representations simultaneously. Their approximate deployed backbone footprints are 808M/421.6G and 857M/532.4G (parameters/FLOPs), respectively, before small fusion-head overhead. These systems are therefore retained as high-compute context rather than direct deployment competitors.

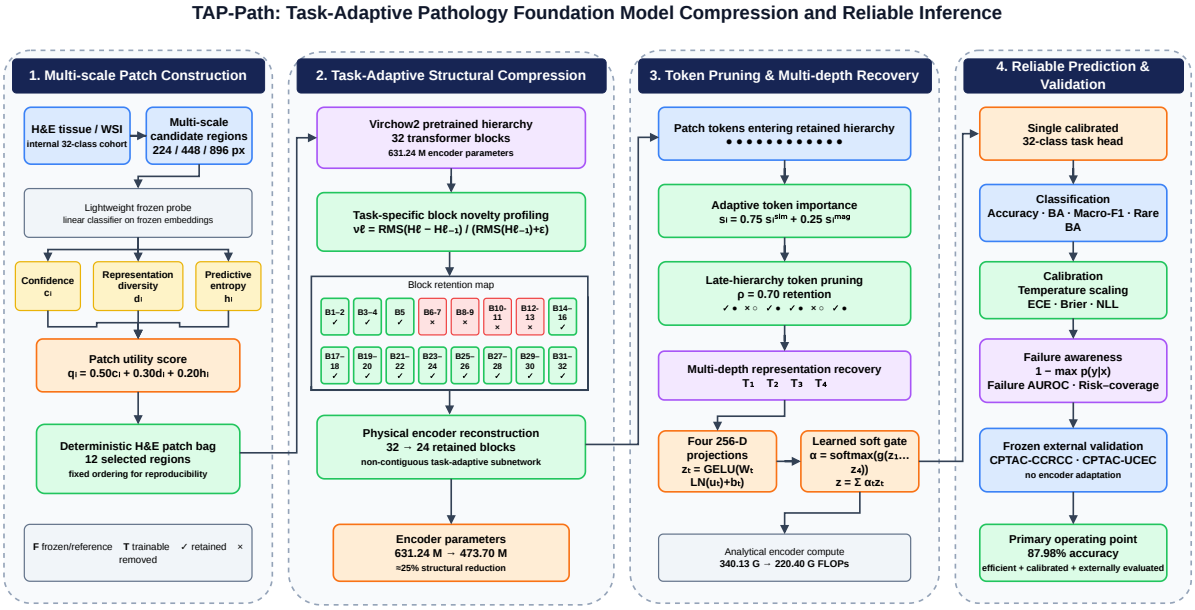


Figure 3: Architecture of the proposed TAP-Path framework. Multi-scale H&E regions are ranked to construct a deterministic 12-patch bag, followed by task-adaptive structural compression of the Virchow2 encoder to the locked one-based block set $\{1,\dots,5\}\cup\{14,\dots,32\}$ (24 of 32 blocks). Adaptive late-hierarchy token pruning retains 70% of informative tokens, while four intermediate feature representations are projected to 256 dimensions and adaptively combined through a learned soft gate before 32-class prediction. The resulting predictions are evaluated for classification performance, calibration, failure awareness, and cross-cohort transfer on CPTAC-CCRCC and CPTAC-UCEC. TAP-Path reduces the encoder from 631.24M to 473.70M parameters and the analytical encoder computation from 340.13 to 220.40 GFLOPs while achieving 87.98% internal test accuracy.

3.5. TAP-Path overview

Let the pretrained encoder contain $L = 32$ transformer blocks,

$$\mathcal{F}(X) = B_L \circ B_{L-1} \circ \cdots \circ B_1(E(X)), \tag{6}$$

where $E(\cdot)$ denotes patch embedding plus prefix-token construction and B_ℓ is transformer block ℓ in Eq. (6). **TAP-Path** transforms $\mathcal{F}$ into a task-specific sparse encoder $\mathcal{F}_{S,\rho}$ parameterized by a retained block set $S \subset \{1,\dots,L\}$ and token-retention ratio ρ. The final model uses $|S| = 24$ and $\rho = 0.70$ after the pruning point.

Figure 3 summarizes the complete TAP-Path workflow from patch construction through structural compression, token pruning, multi-depth recovery, and reliability-oriented evaluation.

The design contains four coupled operations: (i) block novelty profiling and task-adaptive structural selection; (ii) physical removal of unselected transformer blocks; (iii) input-adaptive patch-token pruning; and (iv) gated recovery of features from four depths of the compressed hierarchy. The complete procedure is summarized in Algorithm 1.

3.6. Task-adaptive block selection

Uniform truncation assumes that later blocks are always more expendable or that useful information is distributed monotonically with depth. We instead estimate the normalized residual change induced by each block. For a profiling sample x and block ℓ, let $H_{\ell-1}$ and H_ℓ denote the token tensors immediately before and after the block. We define block novelty as

$$\nu_\ell(x) = \frac{\sqrt{\frac{1}{ND}\left\|H_\ell - H_{\ell-1}\right\|_F^2}}{\sqrt{\frac{1}{ND}\left\|H_{\ell-1}\right\|_F^2} + \epsilon}, \tag{7}$$

where N is the token count, D is the embedding dimension, and $\epsilon = 10^{-8}$ prevents numerical instability. Profiling is restricted to the development subset and excludes internal test and CPTAC records. The novelty statistic itself does not use class labels. Target-task adaptation is introduced by computing the statistic on the target-domain development distribution and selecting candidate subnetworks according to validation performance under the compression constraint. Thus, $\bar{\nu}_\ell$ is used as a computationally inexpensive structural screening signal rather than a supervised attribution measure. The dataset-level score is

$$\bar{\nu}_\ell = \frac{1}{M}\sum_{m=1}^{M} \nu_\ell(x_m). \tag{8}$$

To preserve low-level token formation and high-level semantic consolidation, the first four and final four blocks are anchored. For a target retained depth K, the remaining $K-8$ blocks are selected from the middle hierarchy according to $\bar{\nu}_\ell$ and then restored to their original order. Because transformer blocks preserve a constant hidden width, block excision does not create a dimensional mismatch. If $H_\ell \in \mathbb{R}^{N\times D}$ is the residual stream after a retained block and blocks

$\ell+1, \ldots, j-1$ are removed, the next retained block receives

$$H_j = B_j(H_\ell). \tag{9}$$

The pretrained weights inside retained modules are unchanged; only the computational graph is shortened. The selected 24-block configuration in the locked experiment retained original blocks

$$\mathcal{S}^\star = \{1, \ldots, 5\} \cup \{14, \ldots, 32\}. \tag{10}$$

using one-based indexing for readability. The practical implementation physically replaces the original block list with the retained modules; consequently, omitted blocks no longer contribute stored encoder parameters or forward-pass computation.

Structural candidates were screened against contiguous-depth controls and task-sparse variants. Candidate selection was validation-only and explicitly constrained by compression. The objective can be written as

$$\max_{\mathcal{S}} \quad A_{\text{val}}(\mathcal{S}) \tag{11}$$

$$\text{s.t.} \quad R_P(\mathcal{S}) \geq r_P, \qquad A_{\text{val}}(\mathcal{S}) \geq A^\star_{\text{val}} - \delta, \tag{12}$$

where $R_P = 1 - P_S/P_{\text{full}}$ is parameter reduction, r_P is the required reduction, and δ is the allowed validation tolerance. This formulation avoids choosing the smallest network when its accuracy has already collapsed.

3.7. Adaptive token pruning

After structural selection, token reduction is applied within the compressed encoder. Let $c \in \mathbb{R}^D$ denote the prefix/class token and $p_i \in \mathbb{R}^D$ a patch token. We first normalize both and compute a class-token similarity,

$$s_i^{\text{sim}} = \frac{p_i^\top c}{\|p_i\|_2 \|c\|_2}. \tag{13}$$

To avoid retaining only tokens aligned with the current class token, we add an activation-magnitude term. With per-image normalized magnitude

$$s_i^{\text{mag}} = \frac{\|p_i\|_2 - \mu_{\|p\|}}{\sigma_{\|p\|} + \epsilon}, \tag{14}$$

the fixed development weighting prioritizes semantic alignment while retaining a secondary activation-magnitude cue; it is frozen before test evaluation. The combined token importance is

$$s_i = 0.75\, s_i^{\text{sim}} + 0.25\, s_i^{\text{mag}}. \tag{15}$$

The prefix token is always retained. Among N_p patch tokens, the model keeps

$$K_p = \lceil \rho N_p \rceil. \tag{16}$$

patches with the largest s_i according to Eqs. (15)–(16), where $\rho \in \{1.00, 0.85, 0.70\}$ was evaluated after structural locking. Token pruning begins at retained execution position $\max(1, \text{round}(24 \times 0.55)) = 13$, i.e., after the 13th block in the retained execution sequence. Ranking is recomputed for every patch and introduces no trainable token-router subnetwork.

The analytical compute reduction arises from both fewer blocks and a smaller token sequence. If $\Phi_\ell(N)$ denotes the FLOPs of transformer block ℓ at token count N, the compressed encoder cost is approximated by

$$\Phi_{\text{TAP}} = \sum_{\ell \in \mathcal{S}_{\text{pre}}} \Phi_\ell(N) + \sum_{\ell \in \mathcal{S}_{\text{post}}} \Phi_\ell(K_p + N_{\text{prefix}}), \tag{17}$$

where $\mathcal{S}_{\text{pre}}$ and $\mathcal{S}_{\text{post}}$ denote retained blocks before and after the pruning point.

3.8. Multi-depth feature recovery

Compression can remove intermediate transformations that a downstream classifier would otherwise exploit. We therefore do not classify from only the final compressed token state. Four approximately evenly spaced taps are collected from the retained hierarchy. Four taps balance hierarchical coverage against projection/gating overhead and cached feature dimensionality. The effective token count K_t depends on tap location: taps before the pruning point observe the dense sequence, whereas later taps observe the retained 70% patch sequence. At tap t, the token sequence is normalized and summarized using the prefix token and mean patch token,

$$u_t = \left[\text{LN}(c_t)\,;\, \frac{1}{K_t} \sum_{i=1}^{K_t} \text{LN}(p_{t,i}) \right]. \tag{18}$$

Each u_t is projected to a 256-dimensional task space:

$$z_t = \text{GELU}(W_t\, \text{LN}(u_t) + b_t). \tag{19}$$

Instead of concatenating the four taps and treating them equally, an adaptive gate estimates their sample-specific contribution. Let

$$z_{\text{cat}} = [z_1; z_2; z_3; z_4]. \tag{20}$$

The gate is

$$\boldsymbol{\alpha} = \text{softmax}\left(W_{g2}\ \text{GELU}\left(W_{g1}\, \text{LN}(z_{\text{cat}})\right)\right), \quad \sum_{t=1}^{4} \alpha_t = 1, \tag{21}$$

and the fused feature becomes

$$z = \sum_{t=1}^{4} \alpha_t z_t. \tag{22}$$

The final classifier is a two-layer MLP with LayerNorm, a 512-unit hidden layer, GELU, dropout 0.18, and a 32-class output layer. This single task head contains 5.70M parameters; the deployed single-head model therefore contains 479.40M parameters in total.

3.9. Optimization objectives

For the primary single-head model, class weights are mild rather than fully inverse-frequency weighted:

$$w_c = \frac{\left(\frac{N}{C\,n_c}\right)^{\gamma}}{\frac{1}{C}\sum_{j=1}^{C}\left(\frac{N}{C\,n_j}\right)^{\gamma}}, \qquad \gamma = 0.20, \tag{23}$$

where n_c is the number of training examples in class c. Weighted cross-entropy uses label smoothing $\epsilon_{\text{ls}} = 0.01$. To discourage degenerate single-tap gating, we define gate entropy

$$H(\boldsymbol{\alpha}) = -\sum_{t=1}^{4} \alpha_t \log(\alpha_t + \epsilon) \tag{24}$$

and optimize

$$\mathcal{L}_{\text{primary}} = \mathcal{L}_{\text{CE}} - \lambda_H H(\boldsymbol{\alpha}), \qquad \lambda_H = 0.002. \tag{25}$$

Rare-class sensitivity was investigated after the main architecture was locked. Five validation-only objectives were compared: mild weighted cross-entropy, Balanced Softmax, logit adjustment with $\tau = 0.5$, logit adjustment with $\tau = 1.0$, and class-balanced focal loss. For logit adjustment,

$$\tilde{o}_c = o_c + \tau \log(\pi_c + \epsilon), \tag{26}$$

where o_c is class logit c and π_c is the empirical training prior. This experiment is reported as an ablation of the *same compressed architecture*; the alternative objective is evaluated as a separate operating point from the primary head.

3.10. Training and model selection

The optimization pipeline is separated into architecture selection and task-head learning so that the expensive foundation backbone is not repeatedly fine-tuned for every candidate.

Stage 1: reduced class-aware screening. Structural search uses 1,400 training and 700 validation records selected deterministically with class awareness. Only the first two ranked patches per selected image are processed during screening. This subset exists solely to rank compression candidates efficiently.

Stage 2: structural search. Contiguous-depth controls and non-contiguous task-sparse candidates are constructed from the frozen Virchow2 parent. Each candidate is evaluated using screening features and a lightweight validation probe. The selection criterion jointly considers validation utility and compression; internal test data are not used to choose retained blocks.

Stage 3: token-retention search. After the strongest compressed structures are identified, $\rho \in \{1.00, 0.85, 0.70\}$ is evaluated on validation data. TaskSparse24 with $\rho = 0.70$ is locked before final representation extraction.

Stage 4: full 12-patch feature extraction. The physically reconstructed encoder is run over the complete deterministic 12-patch bags. CUDA automatic mixed precision is enabled where supported. Screening encoder batch size is 4, final feature-extraction batch size is 6, and the development GPU is an NVIDIA GeForce RTX 5060 Ti with approximately 16 GB VRAM.

Stage 5: task-head optimization. The single gated multi-tap head is trained on cached image-level compressed features for at most 130 epochs using AdamW, learning rate 7×10^{-4}, weight decay 2×10^{-4}, batch size 384, gradient clipping at 5, dropout 0.18, and early-stopping patience 18. Three seeds (42, 123, 2026) quantify optimization variability.

Stage 6: calibration and locked evaluation. Temperature scaling is fitted only to validation logits using LBFGS. The locked internal test set is then evaluated. CPTAC-CCRCC and CPTAC-UCEC are processed only after internal configuration locking; no external label, threshold, temperature, or architecture decision modifies TAP-Path.

Structural-screening performance and final test performance are not directly comparable because they correspond to distinct experimental protocols. Screening uses a reduced class-aware subset and two patches per image for efficient candidate ranking, whereas final evaluation uses the complete deterministic 12-patch representation and the trained multi-depth recovery head.

3.11. Common-probe diagnostic

To assess representation quality independently of model-specific task heads, we additionally trained an identical lightweight probe on each frozen representation. The standardized readout produced lower accuracy than the complete TAP-Path model with its multi-depth gated head, indicating that the task-specific recovery module contributes additional downstream discrimination beyond the compressed representation alone. Because the common-probe experiment evaluates representation quality under a standardized classifier rather than the complete TAP-Path inference architecture, it is reported as a complementary diagnostic analysis.

3.12. Reliability metrics

For each trained seed, post-hoc temperature scaling is fitted on validation logits only. Given logits $\boldsymbol{o}$ and temperature $T > 0$,

$$p_c = \frac{\exp(o_c/T)}{\sum_{j=1}^{C}\exp(o_j/T)}. \tag{27}$$

T is optimized by minimizing validation NLL. ECE is computed over 15 confidence bins,

$$\text{ECE} = \sum_{b=1}^{B} \frac{|I_b|}{n}\left|\text{acc}(I_b) - \text{conf}(I_b)\right|. \tag{28}$$

Algorithm 1 TAP-Path: task-adaptive structural and token pruning with multi-depth recovery

Require: Pretrained Virchow2 encoder $\{B_\ell\}_{\ell=1}^{32}$; training set $\mathcal{D}_{tr}$; validation set $\mathcal{D}_{va}$; deterministic candidate-patch manifest
Ensure: Compressed encoder $\mathcal{F}_{\mathcal{S}^\star,\rho^\star}$ and calibrated 32-class task head

Extract frozen Hibou-B candidate-patch embeddings and L2-normalize them
Fit the multinomial logistic patch probe on $\mathcal{D}_{tr}$ only
for each candidate patch i **do**
    Compute probe confidence c_i and entropy h_i using Eqs. (2)–(3)
end for
for each image **do**
    Reserve the highest-confidence patch at every available scale
    Add the two highest-entropy remaining patches
    while fewer than 12 patches are selected **do**
        Compute confidence–diversity–entropy utility using Eq. (1)
        Add the highest-utility remaining patch
    end while
end for
Profile Virchow2 block novelty $\bar{v}_\ell$ on development data using Eq. (7)
Construct contiguous-depth controls and non-contiguous task-sparse block candidates
for each candidate block set $\mathcal{S}$ **do**
    Physically instantiate only blocks in $\mathcal{S}$
    Extract reduced-protocol screening features and fit the fixed lightweight validation probe
    Record validation accuracy, BA, macro-F1, encoder parameters, and analytical FLOPs
end for
Lock the validation-Pareto structural candidate $\mathcal{S}^\star$ (24 of 32 blocks)
for $\rho \in \{1.00, 0.85, 0.70\}$ **do**
    Score tokens with Eq. (15); retain K_p tokens using Eq. (16)
    Evaluate validation performance and analytical compute
end for
Lock $\rho^\star = 0.70$ and rebuild the executable compressed encoder
for seed $\in \{42, 123, 2026\}$ **do**
    Extract the full deterministic 12-patch representation
    Recover four hierarchy taps and compute the adaptive gate using Eq. (21)
    Fuse taps using Eq. (22) and optimize the single 32-class head
    Fit temperature scaling on validation logits only
    Evaluate the locked internal test set and frozen CPTAC cohorts
end for
Report predictive, efficiency, calibration, failure-detection, selective-risk, rare-class, and bootstrap metrics

We additionally report multiclass Brier score

$$\text{Brier} = \frac{1}{n}\sum_{i=1}^{n}\sum_{c=1}^{C}(p_{ic} - y_{ic})^2 \tag{29}$$

and NLL. These measures are complementary: ECE summarizes bin-level calibration, while NLL and Brier are proper scoring rules sensitive to the full predictive distribution.

For failure detection, each example receives an uncertainty score

$$u_i = 1 - \max_c p_{ic}. \tag{30}$$

Ground-truth failures are $e_i = \mathbb{1}[\arg\max_c p_{ic} \neq y_i]$. The AUROC between u_i and e_i measures whether errors tend to receive higher uncertainty. Risk–coverage analysis sorts examples by confidence and evaluates error rate among the retained fraction. This analysis measures whether model confidence can rank prediction errors and support selective retention of higher-confidence samples.

Rare classes are defined from the training distribution using the lower quartile of positive class counts. Rare-class balanced accuracy is the unweighted mean recall over those classes:

$$\text{RareBA} = \frac{1}{|\mathcal{R}|}\sum_{c\in\mathcal{R}}\frac{\text{TP}_c}{\text{TP}_c + \text{FN}_c}. \tag{31}$$

3.13. Statistical analysis

Three independent seeds are reported as mean±SD. Stratified bootstrap confidence intervals are estimated from frozen test predictions using $B = 2000$ resamples:

$$\text{CI}_{95}(m) = \left[Q_{0.025}\{m_b\}_{b=1}^{B}, Q_{0.975}\{m_b\}_{b=1}^{B}\right]. \tag{32}$$

Because aligned prediction vectors are not available for every baseline, marginal confidence intervals are not used as a substitute for a paired superiority test. Small accuracy differences are therefore described as comparable/slightly higher performance, whereas parameter and FLOP reductions are reported as deterministic architectural differences.

3.14. Implementation details

Experiments were implemented in Python 3.11 with PyTorch and the `timm` model library. Foundation models were used as pretrained encoders; mixed-precision feature extraction was enabled on CUDA where supported. The reported development system used an NVIDIA GeForce RTX 5060 Ti GPU. Structural screening used a reduced validation protocol to avoid repeatedly executing all 12 patches for every candidate, whereas the locked winner was re-evaluated under the full 12-patch protocol.

The final single-head task model used four taps, 256-dimensional tap projections, dropout 0.18, up to 130 training epochs, early stopping with patience 18, AdamW with learning rate 7×10^{-4} and weight decay 2×10^{-4}, gradient clipping at 5, and three seeds (42, 123, 2026). The task-head batch size was 384 on cached image-level representations. Temperature scaling used LBFGS on the validation logits. The final runtime profile of the compressed model was 31.85±1.21 ms per image on the development GPU, corresponding to 31.40 images/s under the measured profiling configuration.

4. Results

4.1. Structural compression

The first question was whether selecting blocks by task-dependent novelty provides a better compressed representation than simply truncating the model. Table 3 summarizes the validation-only screening. Full Depth32 achieved the highest absolute screening accuracy (72.96%) but retained the entire encoder. A contiguous Depth24 model reduced encoder parameters by 24.96% but reached 68.28% screening accuracy. At the same parameter count, TaskSparse24 improved screening accuracy to 69.79%, supporting the use of non-contiguous task-adaptive block retention. More aggressive TaskSparse22 and TaskSparse20 configurations reduced parameters by 31.20% and 37.43%, respectively, but incurred additional validation loss.

Table 3: Validation-only structural screening of contiguous-depth and task-adaptive sparse subnetworks. Accuracy and balanced accuracy were obtained with the reduced screening protocol and were used only for architecture selection. Parameter counts and analytical FLOPs quantify the corresponding encoder footprint.

Candidate	Acc. (%)	BA (%)	Params (M)	FLOPs (G)
Depth32	72.96	73.08	631.24	340.13
Depth28	70.39	70.31	552.47	297.66
Depth24	68.28	68.38	473.70	255.19
TaskSparse24	69.79	69.84	473.70	255.19
TaskSparse22	69.18	69.30	434.32	233.96
TaskSparse20	66.47	66.50	394.94	212.73

Figure 4 visualizes the screening frontier. Importantly, the screening accuracies are not directly compared with the final test accuracies because they were obtained with a reduced screening protocol. Their purpose was model selection under equal candidate conditions.

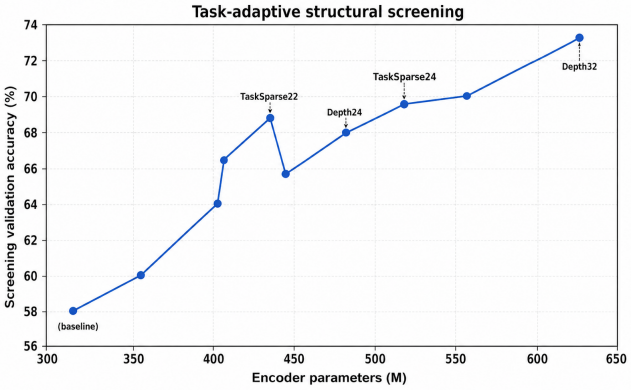


Figure 4: Accuracy–parameter trade-off during validation-only structural screening. Contiguous-depth models provide truncation controls, whereas TaskSparse candidates retain non-contiguous blocks selected from the pretrained hierarchy. TaskSparse24 improves over contiguous Depth24 at the same parameter budget and was retained for subsequent token analysis.

4.2. Token-pruning ablation

Token-retention results are shown in Table 4. For TaskSparse24, retaining 85% of patch tokens increased validation accuracy from 69.79% to 70.39% while reducing analytical FLOPs from 255.19G to 237.71G. Retaining 70% produced the strongest validation score (71.60%) and further reduced compute to 220.40G, a 35.20% reduction relative to full Virchow2. The same 70% ratio was less effective for TaskSparse22, indicating an interaction between structural depth and token sparsity rather than a universal benefit from dropping tokens.

Table 4: Validation-only token-retention analysis after structural screening. Token retention was varied for the two leading sparse structures to assess the interaction between network depth and token sparsity. FLOPs are analytical encoder estimates under each retained-token configuration.

Structure	Tokens (%)	Acc. (%)	BA (%)	FLOPs (G)
TaskSparse24	100	69.79	69.84	255.19
TaskSparse24	85	70.39	70.42	237.71
TaskSparse24	**70**	**71.60**	**71.64**	**220.40**
TaskSparse22	100	69.18	69.30	233.96
TaskSparse22	85	68.73	68.56	218.07
TaskSparse22	70	67.98	67.97	202.33

The validation results indicate that moderate adaptive token pruning can improve the accuracy–compute operating point after stable early representations have formed. This behavior is consistent with removal of redundant or weakly aligned token content and motivates joint selection of structural depth and token retention under the validation protocol. Figure 5 summarizes this depth–token interaction.

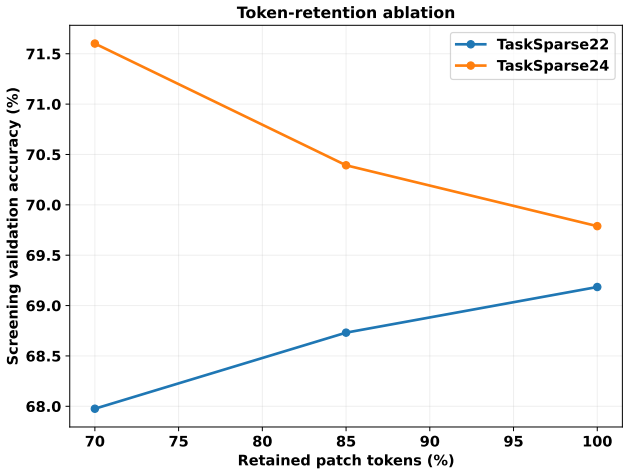


Figure 5: Effect of patch-token retention after structural selection. For TaskSparse24, reducing retention from 100% to 70% improves validation accuracy while reducing analytical encoder FLOPs; the same trend is not observed for TaskSparse22, indicating an interaction between retained depth and token sparsity.

4.3. Foundation-model comparison

Table 5 presents the central same-task comparison. The proposed single-head **TAP-Path** configuration achieved 87.98 ± 0.067% test accuracy across three seeds, compared with 86.89% for full Virchow2 and 87.67% for UNI2-h. The corresponding macro-F1 was 82.38 ± 0.48%, compared with 80.94% for Virchow2 and 81.75% for UNI2-h, while balanced accuracy reached 81.26 ± 0.49%. These results place **TAP-Path** at the highest observed accuracy and macro-F1 among the evaluated single-backbone systems while substantially reducing the computational footprint of the large-model parent. Because paired baseline prediction files were unavailable, the comparison is reported as an observed performance difference without a paired significance test.

The architectural reductions are substantial. Full Virchow2 contains approximately 631–632M encoder parameters, whereas the physically compressed encoder contains 473.70M, corresponding to a 24.96% reduction. Including the task head, the deployed **TAP-Path** configuration contains 479.40M parameters, and analytical encoder FLOPs decrease from 340.13G to 220.40G (35.20%). Relative to UNI2-h, **TAP-Path** also operates with substantially fewer parameters and analytical FLOPs while achieving comparable or higher predictive performance. Hibou-B and CONCH occupy a lower-compute regime, whereas **TAP-Path** defines a favorable operating point within the high-performing large-foundation-model regime.

Table 6 places the high-compute fusion experiments beside the single-backbone systems. These experiments show that comparable discrimination can also be obtained through multi-foundation-model fusion, but at substantially greater deployment-scale parameter and compute requirements. In contrast, TAP-Path achieves its operating point with a single physically compressed backbone.

The corresponding accuracy–efficiency trade-offs are summarized by parameter count and analytical computation. The corrected high-compute fusion references are included in both plots: Virchow2+StaticTriFusion reaches 87.43% accuracy and UNI2-h+DenseTriGate reaches 87.91%, whereas TAP-Path reaches 87.98%. Including the high-compute fusion references further illustrates the resulting Pareto trade-off. TAP-Path achieves 87.98% accuracy using 479.40M deployed parameters and 220.40G analytical encoder FLOPs, while the fusion systems require substantially greater deployment-scale parameters and compute for similar predictive performance.

Figure 6a shows the accuracy–parameter trade-off.

Figure 6b shows the corresponding accuracy–compute trade-off.

4.4. Reliability analysis

Reliability was evaluated from complementary probabilistic and confidence-ranking perspectives. Calibration and probability quality were assessed with ECE, NLL, and Brier score, while failure-detection AUROC and selective risk–coverage quantified whether predictive confidence separated correct and incorrect cases. Rare-class sensitivity is analyzed separately in Section 4.5 to preserve the distinction between the primary task head and the validation-selected rare-aware operating point. Table 7 summarizes the primary-model reliability results. For the primary three-seed TAP-Path task head, ECE is 0.0301 ± 0.0022, NLL is 0.4694 ± 0.0015, Brier score is 0.1800 ± 0.0005, and failure-detection AUROC is 0.9047 ± 0.0060. These measurements indicate that TAP-Path maintains informative confidence estimates despite structural and token pruning. In particular, its Brier score and failure-detection AUROC improve over the full Virchow2 baseline, whereas UNI2-h remains marginally better in ECE and NLL. The reliability claim is therefore based on the joint behavior of several metrics rather than on a single calibration statistic.

Table 7: Reliability comparison for the primary large-model configurations. Lower expected calibration error (ECE) and Brier score are preferred, whereas higher failure-detection AUROC indicates stronger confidence-based separation of correct and incorrect predictions. TAP-Path values are mean±SD across the three primary task-head seeds. Rare-class sensitivity is reported separately in Section 4.5 because it uses a distinct validation-selected operating point.

Model	ECE ↓	Brier ↓	Fail-AUROC ↑
Virchow2	0.0368	0.1882	0.8920
UNI2-h	**0.0297**	0.1825	0.8920
TAP-Path	0.0301±.0022	**0.1800±.0005**	**0.9047±.0060**

Figure 7 evaluates whether confidence can support selective prediction. At 60% internal coverage, TAP-Path retains approximately 99.2% selective accuracy, and risk rises gradually as progressively less-confident cases are accepted. This monotonic behavior is consistent with a useful abstention signal: the model's low-confidence subset is enriched for errors.

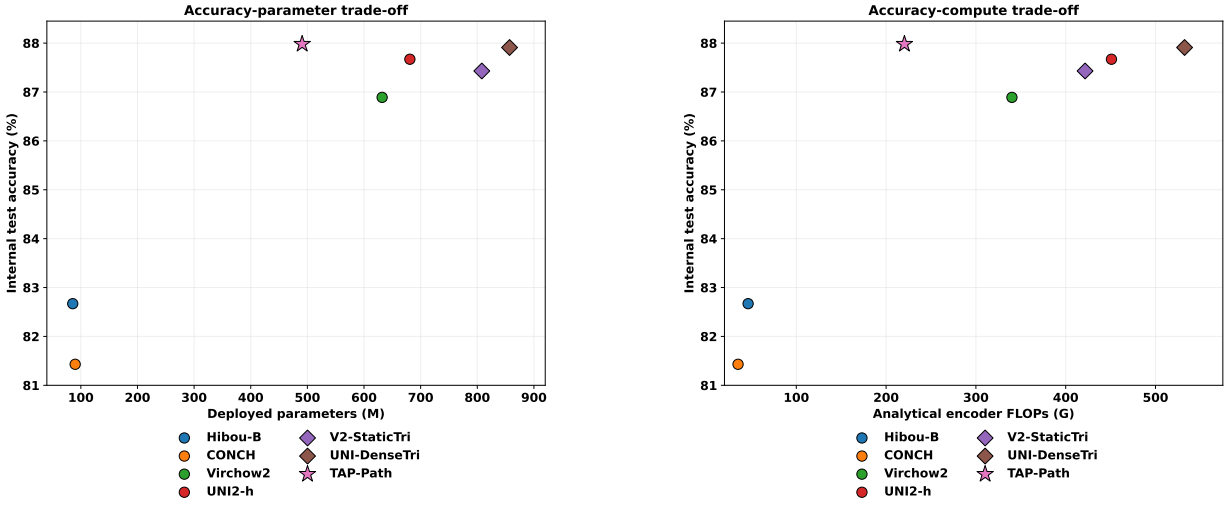
Table 5
Direct comparison of pathology foundation-model encoders on the same 32-class internal test task. **TAP-Path** results are mean±SD across three independently trained task heads. Parameter counts represent the deployed model footprint for **TAP-Path** and the corresponding encoder footprint for reference backbones; analytical encoder FLOPs are reported using a consistent estimation protocol. CPTAC accuracy is shown only as complementary cross-cohort transfer evidence for the two represented classes.

Model	Params	FLOPs	Acc.	BA	F1	CPTAC
Hibou-B	85.7	46.32	82.67	75.17	76.33	**91.53**
CONCH	90.0	35.13	81.43	76.39	76.11	85.76
Virchow2	632.0	340.13	86.89	80.52	80.94	90.76
UNI2-h	681.0	450.97	87.67	81.12	81.75	90.07
TAP-Path (ours)	**479.40**	**220.40**	**87.98±0.067**	**81.26±0.49**	**82.38±0.48**	91.22±0.83

CPTAC accuracy is reported only as complementary transfer evidence for the two represented classes. Hibou-B attains the highest raw CPTAC accuracy, while **TAP-Path** provides the strongest joint internal accuracy–efficiency operating point among the evaluated high-capacity systems.

Table 6
Comparison with higher-compute multi-foundation-model fusion references. Parameter and FLOP values approximate the simultaneously deployed foundation encoders and therefore characterize inference-scale computational requirements rather than the number of trainable task-head parameters.

System	Params	FLOPs	Acc.	F1	Regime
Virchow2	632.0	340.1	86.89	80.94	Full single backbone
UNI2-h	681.0	451.0	87.67	81.75	Full single backbone
Virchow2+StaticTriFusion	~808	~421.6	87.43	81.78	Three-FM fusion
UNI2-h+DenseTriGate	~857	~532.4	87.91	**82.68**	Three-FM fusion
TAP-Path	**479.40**	**220.4**	**87.98**	82.38	**Compressed single backbone**



(a) Accuracy versus deployed parameter count.

(b) Accuracy versus analytical encoder FLOPs.

Figure 6: Accuracy–efficiency trade-offs on the internal test task. (a) TAP-Path occupies a lower-parameter operating point than full Virchow2, UNI2-h, and the higher-compute fusion references while maintaining comparable or higher observed accuracy. (b) TAP-Path reduces Virchow2 analytical encoder computation from 340.13G to 220.40G FLOPs and remains competitive with the higher-compute reference systems.

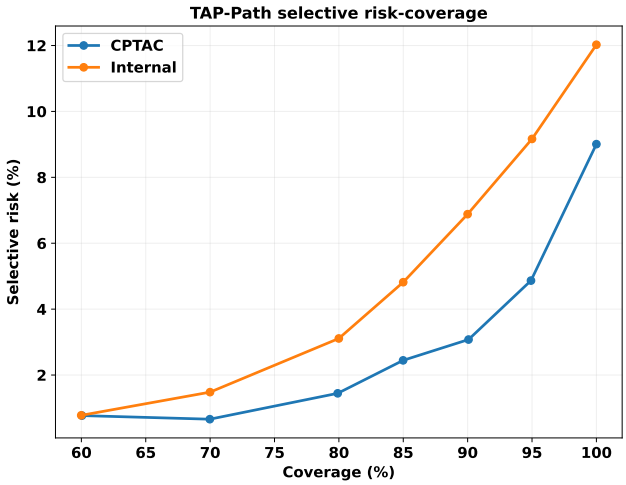


Figure 7: Selective risk–coverage curves for the locked TAP-Path model on the internal test set and pooled CPTAC evaluation. Samples are ordered by predictive confidence; decreasing coverage corresponds to retaining progressively higher-confidence cases, allowing assessment of whether errors concentrate among lower-confidence predictions.

4.5. Rare-class analysis

Rare-class behavior was examined under alternative optimization objectives while keeping the compressed TAP-Path encoder fixed. The primary calibrated configuration is denoted by P, whereas the validation-selected rare-aware configuration is denoted by R. This separation allows the effect of the task-head objective on minority-class sensitivity to be evaluated independently of structural compression.

The rare-aware objectives are reported within the unified ablation analysis in Table 8, preserving the distinction between the primary operating point and the alternative minority-sensitive objectives. On validation data, Balanced Softmax produced the highest rare-class balanced accuracy (70.55%) but reduced overall accuracy. Logit adjustment with $\tau = 1.0$ achieved the highest validation accuracy (90.25%) among the tested objectives while raising rare-class BA from 67.39% for the mild-CE baseline to 69.68%.

When the selected logit-adjusted head was repeated across three seeds on the locked compressed encoder, test rare-class BA increased to 68.64 ± 2.16%, balanced accuracy to 82.29 ± 0.38%, while accuracy decreased to 87.13 ± 0.68%. The comparison demonstrates a controllable operating trade-off within the same compressed encoder: rare-aware optimization increases minority-class sensitivity while shifting the balance among aggregate accuracy, balanced accuracy, and macro-F1. Figure 9 further shows that the overall score is supported by strong discrimination in several well-represented classes: prostate adenocarcinoma reaches F1=0.996, testicular germ-cell tumors F1=0.986, thyroid carcinoma F1=0.986, kidney renal clear-cell carcinoma F1=0.963, and lower-grade glioma F1=0.961. The most difficult classes include rectum adenocarcinoma (F1=0.174), mesothelioma (0.378), esophageal carcinoma (0.439), and cholangiocarcinoma (0.552). The normalized confusion matrix confirms that errors are concentrated in a limited subset of class pairs rather than being uniformly distributed across the 32-class taxonomy. This heterogeneity motivates reporting macro-F1 and rare-class BA alongside top-1 accuracy. The primary result therefore remains the single-head mild-CE configuration in Table 5; the rare-aware result characterizes the minority-sensitivity trade-off of the same compressed encoder.

Figure 8 summarizes the validation trade-off between overall and rare-class performance.

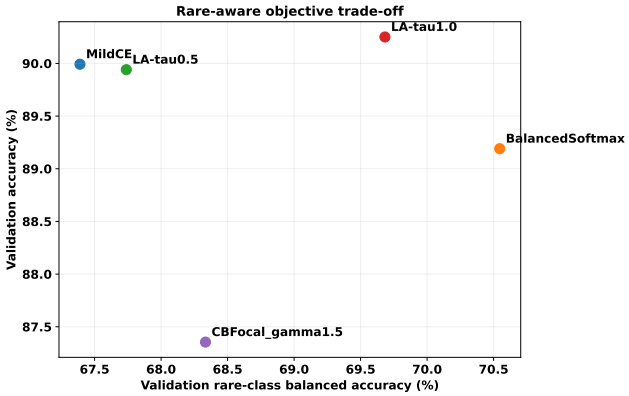


Figure 8: Validation trade-off between overall accuracy and rare-class balanced accuracy for alternative class-imbalance objectives applied to the fixed TAP-Path encoder. The analysis separates the primary mild weighted cross-entropy operating point from the validation-selected logit-adjusted head used for the rare-class sensitivity study.

Figure 9 provides the corresponding class-level F1 profile and normalized confusion matrix.

4.6. Cross-cohort transfer evaluation

Cross-cohort transfer was assessed on two CPTAC disease cohorts. CPTAC-CCRCC contains 209 whole-slide images mapped to the internal kidney renal clear-cell carcinoma class, and CPTAC-UCEC contains 224 whole-slide images mapped to uterine corpus endometrial carcinoma. No CPTAC image or label was used for architecture selection, task-head optimization, or threshold selection.

Across the three primary single-head seeds, TAP-Path achieved 87.40±0.28% accuracy on CCRCC and 94.79±1.44% on UCEC. Pooled over 433 WSIs, accuracy was 91.22±0.83% and present-class balanced accuracy was 91.10±0.81%. Because each cohort corresponds to one mapped target class, cohort-specific recalls/accuracies can be recovered exactly from each frozen seed using

$$\mathrm{BA}_{\mathrm{ext}} = \frac{a_C + a_U}{2}, \tag{33}$$

$$\mathrm{Acc}_{\mathrm{ext}} = \frac{209a_C + 224a_U}{433}. \tag{34}$$

Table 8
Unified ablation of structural depth, token retention, and class-imbalance objectives. Structural and token rows report validation-only screening results obtained with the reduced protocol, whereas loss-function rows use the full locked compressed representation. The final row reports the three-seed test performance of the validation-selected rare-aware head; values from the different stages should therefore be interpreted within their respective evaluation protocols.

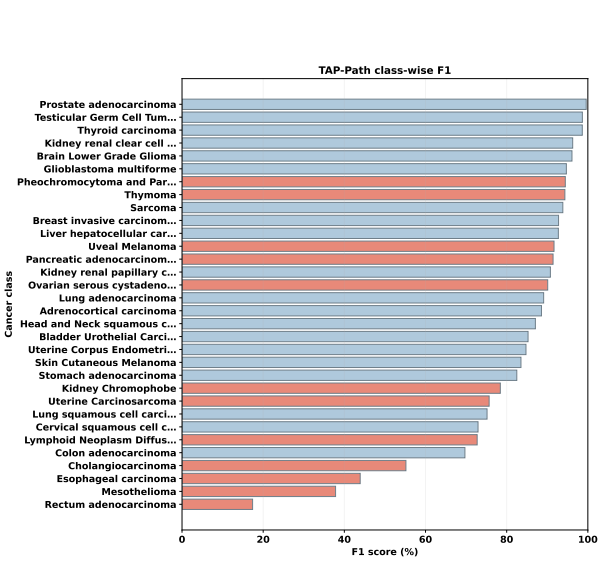
Component	Variant	Acc.	BA	F1	Rare BA	FLOPs
Structure	Depth32	72.96	73.08	71.67	–	340.13
	Depth24	68.28	68.38	67.30	–	255.19
	TaskSparse22	69.18	69.30	68.01	–	233.96
	TaskSparse24	**69.79**	**69.84**	**68.54**	–	255.19
Token ratio	100%	69.79	69.84	68.54	–	255.19
	85%	70.39	70.42	69.10	–	237.71
	70%	**71.60**	**71.64**	**70.32**	–	**220.40**
Loss (validation)	Mild CE	89.99	82.82	84.47	67.39	220.40
	Balanced Softmax	89.19	83.17	81.51	**70.55**	220.40
	Logit adjustment $\tau = .5$	89.94	83.04	84.24	67.74	220.40
	Logit adjustment $\tau = 1$	**90.25**	**83.84**	83.22	69.68	220.40
	CB focal $\gamma = 1.5$	87.35	81.27	81.76	68.33	220.40
Rare-aware 3-seed test	selected $\tau = 1$ head	87.13±.68	82.29±.38	80.74±.65	68.64±2.16	220.40

Table 9: Cross-cohort transfer of TAP-Path to CPTAC clear cell renal cell carcinoma (CCRCC) and uterine corpus endometrial carcinoma (UCEC). Values are mean±SD across the three primary task-head seeds. No CPTAC samples were used for architecture selection, task-head optimization, or threshold selection.

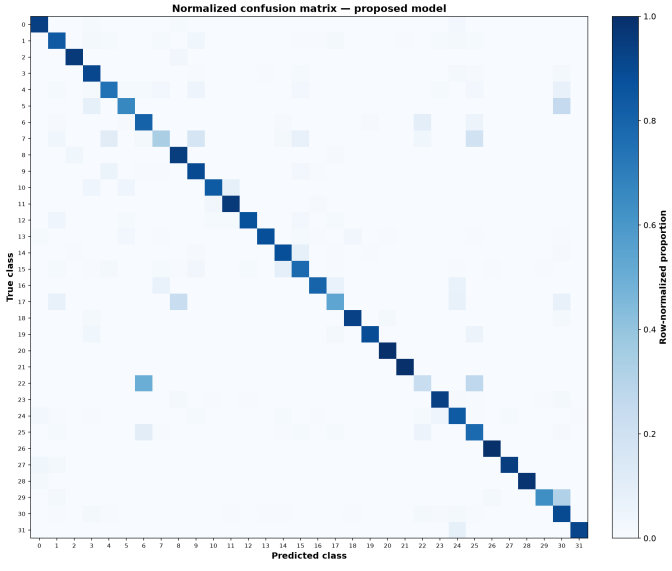
Cohort	WSIs	Accuracy (%)
CPTAC-CCRCC	209	**87.40±0.28**
CPTAC-UCEC	224	**94.79±1.44**
Pooled CPTAC	433	**91.22±0.83**
Present-class BA	433	**91.10±0.81**

Table 9 reports the cohort-specific values. Transfer was stronger for UCEC than CCRCC, supporting disease-specific reporting rather than a pooled score alone. Because only two classes are represented, these results characterize transfer to the evaluated morphologies and do not establish 32-class external generalization.

Across the 433 CPTAC images, **TAP-Path** achieved 91.22 ± 0.83% accuracy and 91.10 ± 0.81% present-class balanced accuracy across the three task-head seeds. ECE was 0.0323±0.0086 and failure-detection AUROC was 0.8974±0.0089, indicating that the locked compressed representation retained useful predictive and confidence-ranking behavior on the evaluated cross-cohort data.



(a) Per-class F1 scores; rare classes are identified from the lower quartile of training frequencies.



(b) Row-normalized 32-class confusion matrix.

Figure 9: Class-level performance of TAP-Path on the locked 32-class internal test set. The per-class F1 profile and row-normalized confusion matrix show that errors are concentrated in a subset of morphologically difficult classes rather than being uniformly distributed across the taxonomy. These views complement aggregate accuracy, balanced accuracy, and macro-F1 by exposing class-specific performance heterogeneity.

Because the CPTAC cohorts contain only two of the 32 internal classes, their raw accuracy is not directly comparable with the internal 32-class task. Among the evaluated baselines, Hibou-B obtained slightly higher CPTAC accuracy (91.53%), while **TAP-Path** exceeded Virchow2 and UNI2-h. The CPTAC experiment is therefore used as complementary cross-cohort evidence rather than as a claim of broad external robustness.

4.7. Deployment efficiency

The final compressed encoder has 473.70M parameters and the single task head has 5.70M, giving 479.40M deployed parameters. Analytical encoder compute is 220.40G FLOPs. An independent profiler supported approximately 213.98G FLOPs for the measured execution path. On the NVIDIA GeForce RTX 5060 Ti development system, mean inference latency was 31.85 ms with 1.21 ms standard deviation, corresponding to 31.40 images/s. These measurements are hardware-specific, but they confirm that the structural and token changes yield an executable compressed model rather than only a theoretical masking scheme.

4.8. Bootstrap uncertainty

Figure 10 summarizes the resulting uncertainty. The 2,000-resample stratified bootstrap gives approximate 95% intervals of 0.870–0.889 for accuracy, 0.793–0.831 for balanced accuracy, 0.805–0.839 for macro-F1, 0.568–0.698 for rare-class balanced accuracy, and 0.018–0.032 for ECE. The wider rare-class interval reflects the smaller effective support of low-frequency classes.

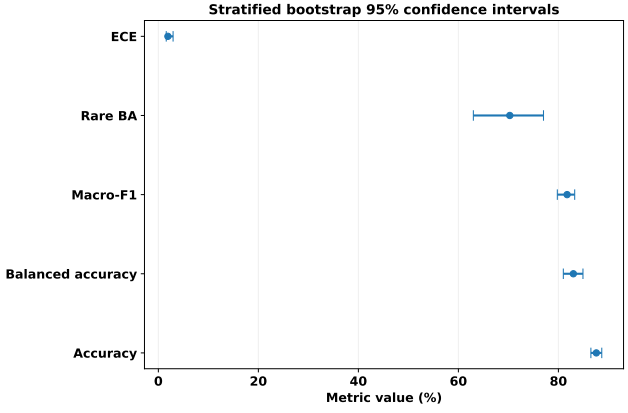


Figure 10: Stratified bootstrap 95% confidence intervals from 2,000 resamples of the locked internal test predictions. Intervals are shown for accuracy, balanced accuracy, macro-F1, rare-class balanced accuracy, and ECE; the wider rare-class interval reflects the smaller effective support of low-frequency classes.

5. Discussion

The main finding of this study is that downstream-efficient inference can be obtained by restructuring the pretrained PFM itself rather than replacing it with a student or retaining the full parent. TAP-Path removes approximately one quarter of the Virchow2 encoder parameters and more than one third of its analytical encoder FLOPs while maintaining competitive 32-class discrimination. The ablations further show that structural and token sparsity are coupled: non-contiguous block selection is preferable to naive truncation at the same parameter budget, and token reduction is beneficial for the selected 24-block structure but not for the more aggressively compressed 22-block variant.

5.1. Task-specific structural compression of pathology foundation models

The comparison between contiguous Depth24 and TaskSparse24 provides evidence that downstream-relevant transformations are distributed non-uniformly across the pretrained hierarchy. Both configurations retain 24 blocks and therefore have the same encoder parameter count, but the non-contiguous TaskSparse24 structure achieves higher validation performance under the common screening protocol. This result extends prior observations of redundancy in transformer components [1, 27] to complete blocks of a large pathology foundation encoder and indicates that uniform depth truncation can discard transformations that remain useful for the target distribution.

The block-novelty criterion measures normalized representational change rather than direct causal importance. A block with a relatively small residual change may still perform useful feature refinement. TAP-Path therefore uses the novelty score to construct candidate subnetworks and relies on target-task validation performance under an explicit compression constraint to select the operating point. Gradient-, Hessian-, or intervention-based criteria could provide complementary estimates of structural importance in future work.

5.2. Interaction between structural and token pruning

The token-retention experiments show that structural and token sparsity are not independent. For TaskSparse24, reducing token retention from 100% to 85% and then 70% improves validation performance while progressively lowering analytical FLOPs. In contrast, the same reductions decrease performance for TaskSparse22. This interaction suggests that token removal is better tolerated when sufficient transformation depth remains available to consolidate local evidence before the sequence is shortened.

This behavior is particularly relevant to histopathology, where a small tissue region may contain diagnostically important morphology. Generic token-efficiency methods demonstrate that substantial spatial redundancy is present in vision transformers [5, 21, 33, 44], but the amount of removable information depends on both task and representation stage. The 70% retention ratio identified here should therefore be interpreted as a validation-selected operating point for the evaluated backbone and task rather than a universal pruning ratio.

5.3. Multi-depth representation recovery

Physical block removal shortens the sequence of transformations available to the downstream classifier. TAP-Path

addresses this by collecting four representations from the retained hierarchy and combining them through a learned gate. The common-probe diagnostic shows that the compressed representation remains informative under an identical lightweight readout, while the multi-depth task head provides additional predictive performance. These findings suggest that information useful for downstream classification remains distributed across the retained hierarchy after compression and can be recovered without restoring the removed blocks.

The multi-depth head introduces additional task-specific parameters, but its 5.70M-parameter footprint is small relative to the reduction in the parent encoder. Future work could jointly optimize tap locations with structural selection or investigate whether simpler fusion rules can preserve the same benefit.

5.4. Computational efficiency at whole-slide scale

The reported FLOP reduction is measured per encoder invocation, whereas WSI analysis repeatedly applies the patch encoder. If a slide contributes N_{WSI} encoded patches, the encoder computation can be approximated by

$$\Phi_{\mathrm{slide}} \approx N_{\mathrm{WSI}} \Phi_{\mathrm{patch}}. \tag{35}$$

For 10,000 encoded patches, reducing analytical cost from 340.13G to 220.40G FLOPs corresponds to approximately 1.20×10^{15} fewer floating-point operations before slide-level aggregation. This calculation illustrates how moderate per-patch savings can accumulate in WSI pipelines. The estimate does not include tissue detection, slide I/O, patch extraction, batching, or downstream aggregation and therefore should not be interpreted as end-to-end clinical latency.

Measured image-level latency provides complementary evidence that the structural changes produce an executable compressed graph rather than only masked computation. On the development GPU, TAP-Path required 31.85 ± 1.21 ms per image under the reported profiling configuration. Direct deployment comparisons across hardware would additionally require standardized precision, batching, kernel, memory, and full-slide preprocessing settings.

5.5. Reliability after compression

Compression can alter predictive confidence even when classification accuracy is maintained. TAP-Path was therefore evaluated using ECE, NLL, Brier score, failure-detection AUROC, and selective risk–coverage in addition to class-balanced performance. The compressed model obtained a lower Brier score and higher failure-detection AUROC than the full Virchow2 baseline, while UNI2-h achieved slightly lower ECE and NLL. Thus, the observed efficiency gain was not accompanied by a uniform degradation of the evaluated reliability measures.

The selective-prediction analysis provides an additional view of confidence quality. Prediction risk decreases as lower-confidence samples are withheld, indicating that errors are concentrated toward the low-confidence end of the ranking. This behavior is relevant to settings in which model confidence may be used to prioritize manual review, although prospective workflow studies would be required to determine clinically appropriate operating thresholds.

Rare-class analysis identifies a related trade-off. The primary objective achieves the strongest aggregate performance, whereas the validation-selected logit-adjusted head increases rare-class balanced accuracy with a modest reduction in overall accuracy. The two operating points demonstrate that minority-class sensitivity can be modified without changing the compressed encoder. They also motivate future pruning criteria that incorporate class-frequency or class-conditional utility during structural and token selection.

5.6. Cross-cohort transfer

The locked TAP-Path configuration was applied to CPTAC-CCRCC and CPTAC-UCEC without CPTAC-specific adaptation or threshold tuning. Performance differed across the two cohorts, with higher accuracy on UCEC than CCRCC, and the pooled result remained competitive with the evaluated foundation-model baselines. Hibou-B achieved slightly higher raw CPTAC accuracy, consistent with prior PFM benchmarks showing that relative model ranking can vary across tasks and cohorts [3, 7, 29].

The CPTAC analysis covers only two classes from the 32-class internal taxonomy and should therefore be interpreted as limited cross-cohort transfer evidence. Broader claims about pan-cancer generalization require evaluation across additional diseases, institutions, scanners, and staining protocols.

5.7. Relation to efficient pathology adaptation methods

Recent pathology-efficiency methods target different computational bottlenecks. PAMT combines representative patch sampling, prompts, and adapters to reduce downstream adaptation cost [22]; SUDA transfers knowledge from a large foundation model to a compact student [46]; and foundation-model/MIL studies examine how patch representations interact with slide-level aggregation [40]. Lightweight architectures instead reduce complexity through model design [15, 18]. TAP-Path addresses a complementary setting by modifying the pretrained parent itself after pretraining and physically removing task-redundant blocks and tokens from its inference pathway.

These strategies can be viewed as complementary stages of efficient PFM deployment. PEFT reduces optimization burden, input selection limits the number of regions presented to the model, in-place pruning reduces the parent inference graph, and distillation can provide further compression when a separate student is acceptable. The present results isolate the contribution of in-place structural and token sparsification and show that this strategy can provide a favorable accuracy–efficiency operating point while retaining measurable reliability under the evaluated conditions.

6. Limitations

Several limitations should be considered. First, the final TAP-Path model contains 479.40M deployed parameters. Although substantially smaller than the parent Virchow2 and UNI2-h systems, it remains larger than compact encoders such as Hibou-B and CONCH. The present contribution therefore targets improved efficiency within the high-capacity PFM regime rather than edge-scale deployment.

Second, analytical FLOPs provide an architecture-level estimate and the reported latency is specific to the development hardware and profiling configuration. End-to-end WSI processing also depends on slide reading, tissue detection, patch extraction, storage I/O, batching, numerical precision, and downstream aggregation. Broader systems evaluation should include full-slide latency, peak memory, energy consumption, and throughput across multiple hardware platforms.

Third, the available cross-cohort evaluation is limited to 433 CPTAC samples from CCRCC and UCEC, representing only two classes from the internal 32-class taxonomy. These results should not be interpreted as pan-cancer external validation. Broader assessment across additional diseases, institutions, scanners, and staining protocols remains necessary.

Fourth, the block-novelty statistic measures normalized representational change and does not directly quantify causal task importance. Alternative criteria based on gradients, Hessian approximations, interventions, or class-conditional utility may identify different subnetworks and could be incorporated into the same validation-constrained selection framework.

Fifth, the rare-aware analysis modifies the task-head objective after the compressed encoder has been locked. Although this analysis shows that minority-class sensitivity can be shifted within the same representation, it does not establish that the structural pruning rule is neutral with respect to class frequency. Incorporating minority-class utility directly into structural and token selection is an important direction for future work.

Sixth, the reliability evaluation is retrospective and model-centered. Calibration, proper scoring rules, failure detection, selective prediction, bootstrap uncertainty, and the limited CPTAC transfer analysis characterize important aspects of predictive reliability, but reader interaction, clinical decision utility, demographic subgroup fairness, adversarial robustness, formal uncertainty guarantees, and prospective operating thresholds were not evaluated.

Finally, the structural selection experiments focus on one large parent PFM and one downstream pan-cancer classification setting. The retained block pattern and 70% token ratio may differ for other parent architectures or endpoints such as segmentation, biomarker prediction, and survival analysis. Evaluating the selection procedure across models and tasks will be necessary to determine which compression patterns generalize.

7. Conclusion

We proposed **TAP-Path**, a task-adaptive framework for in-place compression of pathology foundation models through structural and token pruning. Starting from Virchow2, the selected configuration retains 24 of 32 transformer blocks and 70% of patch tokens after the pruning point, while four retained-depth representations are combined through a lightweight gated head. The compressed encoder reduces parameters by 24.96% and analytical FLOPs by 35.20% while achieving 87.98±0.067% internal test accuracy and 82.38±0.48% macro-F1 across three task-head seeds.

The results indicate that the computation required for a specific downstream pathology task can be smaller than the full pretrained inference hierarchy. Structural and token sparsity should be selected jointly because their effects are coupled, and reliability measures provide information that is not captured by accuracy and computational cost alone. The CPTAC experiment provides complementary cross-cohort evidence for two represented disease classes, but broader multi-institutional validation remains future work. Overall, TAP-Path establishes a practical accuracy–efficiency operating point by compressing the pretrained parent directly while preserving competitive discrimination and informative confidence behavior.

Data Availability

The histopathology data analyzed in this study were derived from publicly available resources from The Cancer Genome Atlas (TCGA) [35] and the Clinical Proteomic Tumor Analysis Consortium (CPTAC) [10, 11]. TCGA data are accessible through the NCI Genomic Data Commons, while the corresponding CPTAC resources are available through NCI-supported data repositories. No cross-cohort evaluation data were used for architecture selection or model optimization.

Declaration of Competing Interest

The authors declare no competing interests.

A. Reproducibility details

A.1. Primary training configuration

Table 10: Primary training and calibration configuration for the locked TAP-Path task head. Hyperparameters were fixed before internal test and CPTAC transfer evaluation.

Setting	Value
Parent foundation model	Virchow2
Retained blocks	24/32
Patch-token retention	70%
Multi-depth taps	4
Tap projection	256
Dropout	0.18
Optimizer	AdamW
Learning rate	7×10^{-4}
Weight decay	2×10^{-4}
Batch size	384
Maximum epochs	130
Early stopping	18 epochs
Gradient clip	5
Label smoothing	0.01
Seeds	42, 123, 2026
Calibration	validation-only temperature scaling

References

[1] An, et al., 2025. Sparse structure exploration and re-optimization for vision transformer, in: Proceedings of the 41st Conference on Uncertainty in Artificial Intelligence, PMLR. pp. 111–131.

[2] Azad, R., et al., 2024. Advances in medical image analysis with vision transformers: A comprehensive review. Medical Image Analysis 91, 103000. doi:10.1016/j.media.2023.103000.

[3] Bareja, R., Carrillo-Perez, F., Zheng, Y., Pizurica, M., Nandi, T.N., Tian, L., Shen, J., Madduri, R., Gevaert, O., 2026. A benchmark study of vision and pathology foundation models for computational pathology. Nature Communications doi:10.1038/s41467-026-76004-6.

[4] Bergner, et al., 2025. Token cropr: Faster vits for quite a few tasks, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 9740–9750.

[5] Bolya, D., Fu, C.Y., Dai, X., Zhang, P., Feichtenhofer, C., Hoffman, J., 2023. Token merging: Your vit but faster, in: International Conference on Learning Representations.

[6] Boudissa, S., Debsarkar, S.S., Kawanaka, H., Aronow, B.J., Prasath, V.B.S., 2025. Vision transformers for histopathological image classification with efficient head pruning. Procedia Computer Science 270, 2274–2283. doi:10.1016/j.procs.2025.09.348.

[7] Campanella, G., Chen, S., Singh, M., et al., 2025. A clinical benchmark of public self-supervised pathology foundation models. Nature Communications 16, 3640. doi:10.1038/s41467-025-58796-1.

[8] Chen, R.J., Chen, C., Li, Y., et al., 2022. Scaling vision transformers to gigapixel images via hierarchical self-supervised learning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 16144–16155.

[9] Chen, R.J., Ding, T., Lu, M.Y., et al., 2024. Towards a general-purpose foundation model for computational pathology. Nature Medicine 30, 850–862. doi:10.1038/s41591-024-02857-3.

[10] Clark, D.J., Dhanasekaran, S.M., Petralia, F., Pan, J., Song, X., Hu, Y., da Veiga Leprevost, F., Reva, B., Lih, T.M., Chang, H.Y., et al., 2019. Integrated proteogenomic characterization of clear cell renal cell carcinoma. Cell 179, 964–983.e31. doi:10.1016/j.cell.2019.10.007.

[11] Dou, Y., Kawaler, E.A., Cui Zhou, D., Gritsenko, M.A., Huang, C., Blumenberg, L., Karpova, A., Petyuk, V.A., Savage, S.R., Satpathy, S., et al., 2020. Proteogenomic characterization of endometrial carcinoma. Cell 180, 729–748.e26. doi:10.1016/j.cell.2020.01.026.

[12] Geifman, Y., El-Yaniv, R., 2019. Selectivenet: A deep neural network with an integrated reject option, in: Proceedings of the 36th International Conference on Machine Learning, PMLR. pp. 2151–2159.

[13] Guo, C., Pleiss, G., Sun, Y., Weinberger, K.Q., 2017. On calibration of modern neural networks, in: Proceedings of the 34th International Conference on Machine Learning, PMLR. pp. 1321–1330.

[14] Huang, Y., Zhao, W., Zhang, Z., et al., 2025. Knowledge-guided adaptation of pathology foundation models effectively improves cross-domain generalization and demographic fairness. Nature Communications doi:10.1038/s41467-025-66300-y.

[15] Islam, M.F., Reza, M.T., Manab, M.A., et al., 2025. Involution-based efficient autoencoder for denoising histopathological images with enhanced hybrid feature extraction. Computers in Biology and Medicine 192, 110174. doi:10.1016/j.compbiomed.2025.110174.

[16] Jahanifar, M., Raza, S.E.A., et al., 2025. Domain generalization in computational pathology: Survey and guidelines. ACM Computing Surveys 57. doi:10.1145/3729524.

[17] Komen, J., de Jong, E.D., Hense, J., et al., 2026. Towards robust foundation models for digital pathology. Nature Communications 17, 5218. doi:10.1038/s41467-026-73923-2.

[18] Kuang, et al., 2025. Lw-ctrans: A lightweight hybrid network of cnn and transformer for 3d medical image segmentation. Medical Image Analysis 102, 103545. doi:10.1016/j.media.2025.103545.

[19] Lee, J., Lim, J., Byeon, K., Kwak, J.T., 2025. Benchmarking pathology foundation models: Adaptation strategies and scenarios. Computers in Biology and Medicine 190, 110031. doi:10.1016/j.compbiomed.2025.110031.

[20] Li, D., Wan, G., Wu, X., Wu, X., He, Y., Chen, Z., Nirmal, A.J., Lian, C.G., Sorger, P.K., Semenov, Y.R., Zhao, C., 2026. A survey on computational pathology foundation models: datasets, adaptation strategies, and evaluation tasks. Knowledge and Information Systems 68, 209. doi:10.1007/s10115-026-02806-1.

[21] Liang, Y., Ge, C., Tong, Z., Song, Y., Wang, J., Xie, P., 2022. Evit: Expediting vision transformers via token reorganizations, in: International Conference on Learning Representations.

[22] Lin, Y., Zhu, Z., Cheng, K.T., Chen, H., 2026. Prompt-guided foundation model tuning for pathology image classification. Medical Image Analysis 113, 104214. doi:10.1016/j.media.2026.104214.

[23] Liu, et al., 2024. Revisiting token pruning for object detection and instance segmentation, in: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision, pp. 2658–2668.

[24] Lu, M.Y., Chen, B., Williamson, D.F.K., et al., 2024. A visual-language foundation model for computational pathology. Nature Medicine 30, 863–874. doi:10.1038/s41591-024-02856-4.

[25] Ma, J., Guo, Z., Zhou, F., et al., 2026. A generalizable pathology foundation model using a unified knowledge distillation pretraining framework. Nature Biomedical Engineering 10, 545–564. doi:10.1038/s41551-025-01488-4. published online 2025.

[26] Marchetti, et al., 2025. Efficient token pruning in vision transformers using an attention-based multilayer network. Expert Systems with Applications 279, 127449. doi:10.1016/j.eswa.2025.127449.

[27] Michel, P., Levy, O., Neubig, G., 2019. Are sixteen heads really better than one?, in: Advances in Neural Information Processing Systems.

[28] Nechaev, D., Pchelnikov, A., Ivanova, E., 2024. Hibou: A family of foundational vision transformers for pathology. arXiv preprint arXiv:2406.05074 .

[29] Neidlinger, P., El Nahhas, O.S.M., Muti, H.S., et al., 2026. Benchmarking foundation models as feature extractors for weakly supervised computational pathology. Nature Biomedical Engineering 10, 1113–1123. doi:10.1038/s41551-025-01516-3. published online 2025.

[30] Nicke, T., Schaefer, J.R., Hoefener, H., et al., 2025. Tissue concepts: supervised foundation models in computational pathology. Computers in Biology and Medicine 186, 109621. doi:10.1016/j.compbiomed.2024.109621.

[31] Oquab, M., Darcet, T., Moutakanni, T., et al., 2024. Dinov2: Learning robust visual features without supervision. Transactions on Machine

Learning Research .

[32] Ovadia, Y., Fertig, E., Ren, J., et al., 2019. Can you trust your model's uncertainty? evaluating predictive uncertainty under dataset shift, in: Advances in Neural Information Processing Systems.

[33] Rao, Y., Zhao, W., Liu, B., Lu, J., Zhou, J., Hsieh, C.J., 2021. Dynamicvit: Efficient vision transformers with dynamic token sparsification, in: Advances in Neural Information Processing Systems.

[34] Silva-Rodriguez, J., et al., 2025. Towards foundation models and few-shot parameter-efficient fine-tuning for volumetric organ segmentation. Medical Image Analysis 103. doi:10.1016/j.media.2025.103596.

[35] The Cancer Genome Atlas Research Network, Weinstein, J.N., Collisson, E.A., Mills, G.B., Shaw, K.R.M., Ozenberger, B.A., Ellrott, K., Shmulevich, I., Sander, C., Stuart, J.M., 2013. The cancer genome atlas pan-cancer analysis project. Nature Genetics 45, 1113–1120. doi:10.1038/ng.2764.

[36] Vorontsov, E., Bozkurt, A., Casson, A., et al., 2024. A foundation model for clinical-grade computational pathology and rare cancers detection. Nature Medicine 30, 2924–2935. doi:10.1038/s41591-024-03141-0.

[37] Wang, et al., 2022. Transformer-based unsupervised contrastive learning for histopathological image classification. Medical Image Analysis 81, 102559. doi:10.1016/j.media.2022.102559.

[38] Wang, X., Zhao, J., Marostica, E., et al., 2024. A pathology foundation model for cancer diagnosis and prognosis prediction. Nature 634, 970–978. doi:10.1038/s41586-024-07894-z.

[39] Xu, et al., 2024. Gtp-vit: Efficient vision transformers via graph-based token propagation, in: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision, pp. 86–95.

[40] Xu, et al., 2025. When multiple instance learning meets foundation models: Advancing histological whole slide image analysis. Medical Image Analysis 101, 103456. doi:10.1016/j.media.2025.103456.

[41] Xu, H., Usuyama, N., Bagga, J., et al., 2024. A whole-slide foundation model for digital pathology from real-world data. Nature 630, 181–188. doi:10.1038/s41586-024-07441-w.

[42] Yan, F., Wu, J., Li, J., et al., 2025. Pathorchestra: A comprehensive foundation model for computational pathology with over 100 diverse clinical-grade tasks. arXiv preprint arXiv:2503.24345 .

[43] Yang, Z., Wei, T., Liang, Y., et al., 2025. A foundation model for generalizable cancer diagnosis and survival prediction from histopathological images. Nature Communications 16. doi:10.1038/s41467-025-57587-y.

[44] Yin, H., Vahdat, A., Alvarez, J.M., Mallya, A., Kautz, J., Molchanov, P., 2022. A-vit: Adaptive tokens for efficient vision transformer, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 10809–10818. doi:10.1109/CVPR52688.2022.01054.

[45] Zhang, Metaxas, D., 2024. On the challenges and perspectives of foundation models for medical image analysis. Medical Image Analysis 91, 102996. doi:10.1016/j.media.2023.102996.

[46] Zhong, L., Qian, K., Zhao, W., et al., 2026. Suda: Simultaneous unsupervised knowledge distillation and adaptation of foundation models for efficient pathological image analysis. Medical Image Analysis 113, 104177. doi:10.1016/j.media.2026.104177.